PHYSICAL CHEMISTRY

GUIDED REVISION PLAN-SCORE ADVANCED

Topic : Solid State (Question Bank # 10)

MM : 70 **Time : 45 Min.**

SECTION–I : (i) Only One option correct Type

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. It was found that in 30 unit cells of ZnS, four atoms are of trivalent metal. How many vacant tetrahedral voids are present in 30 unit cells of ZnS?

(A) 2 (B) 62 (C) 122 (D) 126

2. A solid element (monoatomic) exist as cubic crystal. If its atomic radius is 1.0 Å and the ratio of packing fraction and density is 0.1 cm^3/gm, then the atomic mass of the element is ($N_A = 6 \times 10^{23}$)

(A) 8π (B) 16π (C) 80π (D) 4π

3. A metal crystallises in such a lattice that only 70% of the total space crystal is occupied by the atoms. If the atomic mass of metal is 32π gm mol^{-1} and atomic radius is 0.2 nm, the density of the metal is :

(A) 7.0 gm/cm^3 (B) 3.5 gm/cm^3

(C) 10.5 gm/cm^3 (D) 14.0 gm/cm^3

4. Cesium iodide consists of a simple cubic lattice of I^- ions with Cs^+ ions in the cubic holes. If the cell edge length is 0.445 nm, what is the Cs–I interatomic distance ?

(A) 0.223 nm (B) 0.314 nm

(C) 0.385 nm (D) 0.770 nm

5. The ZnS (zinc blende) structure is cubic. The unit cell may be described as a face-centered sulfide ion sublattice with zinc ions in the centers of alternating minicubes made by partitioning the main cube into 8 equal parts. How many nearest neighbors does each Zn^{2+} have and what angle is made by the lines connecting any Zn^{2+} to any two of its nearest neighbors?

(A) 4 , 90° (B) 6 , 90°

(C) 4 , 109.5° (D) 6 , 109.5°

6. Which of the following oxides behaves as conductor or insulator depending upon temperature?

(A) TiO (B) SiO_2 (C) TiO_3 (D) MgO

7. Which of the following oxides shows electrical properties like metals?

(A) SiO_2 (B) MgO (C) $SO_2(s)$ (D) CrO_2

8. A compound has cubical unit cell in which X atoms are found at 6 corners and Y atoms are present at remaining corners and at face center which are not opposite to each other and Z-atoms are present at body center and remaining face center. Calculate density in amu/$(A°)^3$, if atomic mass of X, Y and Z are 20, 40 and 60 amu respectively and side of unit cell (2Å).

(A) $\frac{235}{8}$ (B) $\frac{940}{8}$ (C) $\frac{470}{8}$ (D) $\frac{235}{16}$

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(ii) One or more options correct Type

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

9. The incorrect statement(s) for the packing of identical spheres in two dimension is :

(A) For square close packing , coordination number is 4.

(B) For hexagonal close packing, coordination number is 6.

(C) There is only one void per atom in both, square and hexagonal close packing.

(D) Hexagonal close packing is more efficiently packed than square close packing.

10. Which of the following statements are correct?

(A) Ferrimagnetic substances lose ferrimagnetism on heating and become paramagnetic.

(B) Ferrimagnetic substances do not lose ferrimagnetism on heating and remain ferrimagnetic.

(C) Antiferromagnetic substances have domain structures similar to ferromagnetic substances and their magnetic moments are not cancelled by each other.

(D) In ferromagnetic substances all the domains get oriented in the direction of magnetic field and remain as such even after removing magnetic field.

11. If observed density of crystal is less than ideal density, then the defect may be -

(A) Schottky defect (B) Frenkel defect

(C) substitutional impurity (D) metal deficiency defect

12. The co-ordination number of FCC structure for metals is 12, since

(A) each atom touches 4 others in same plane, 3 in plane above and 3 in plane below.

(B) each atom touches 4 others in same plane, 4 in plane above and 4 in plane below.

(C) each atom touches 6 others in same layer, 3 in layer above and 3 in layer below.

(D) each atom touches 3 others in same layer, 6 in layer above and 6 in layer below.

(iii) Paragraph Type

This section contains **1 paragraph** , describing theory, experiment, data etc. **2 questions** relate to one paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Q. 13 & Q. 14

Spinel is an important class of oxide having two types of metal ions. A type of spinal have ccp arrangement to O^{2-} ions in which Fe^{2+} cation occupy 1/8th of tetrahedral voids and Fe^{3+} cation occupies half of the octahedral voids. If '**a**' be the edge of unit cell then.

13. What is formula of spinel -

(A) Fe_2O_3 (B) Fe_3O_4 (C) FeO (D) Fe_4O_5

14. What is the minimum internuclear distance between Fe^{2+} ion & Fe^{3+} ion in spinel -

(A) $\frac{\sqrt{3}a}{4}$ (B) $\frac{a}{2}$ (C) $\frac{a}{\sqrt{2}}$ (D) a

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(iv) Matching List Type

This Section contains **1 multiple choice questions. Each question has matching lists.** The codes for the lists. have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(0)**

15.

List-I Type of ionic compound	**List-II**
(P) NaCl type	(A) Cation at 50% of tetrahedral void
(Q) ZnS type	(B) Cation at 100% of tetrahedral void
(R) CsCl type	(C) Cation at cubic void
(S) Na_2O type	(D) Cation at 100% octahedral void

Code :

	P	**Q**	**R**	**S**
(A)	4	1	3	2
(B)	2	1	4	1
(C)	1	2	3	4
(D)	4	3	4	2

(v) Table Type

- This section contains **1** question of matching type.
- This section contains **1** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

Column-I	**Column-II**	**Column-III**
(A) Hexagonal	(P) $a = b \neq c$	(I) Primitive
(B) Orthorhombic	(Q) $a \neq b \neq c$	(II) Body centred
(C) Monoclinic	(R) $\alpha = \beta \neq \gamma$	(III) Face centered
(D) Triclinic	(S) $\alpha \neq \beta \neq \gamma$	(IV) End centered

16. Which of the following is only correct match :
(A) A, Q, II (B) C, R, II (C) D, S, II (D) A, P, I

17. Which of the following is only incorrect match :
(A) A, P, I (B) B, P, IV (C) C, R, I (D) D, S, I

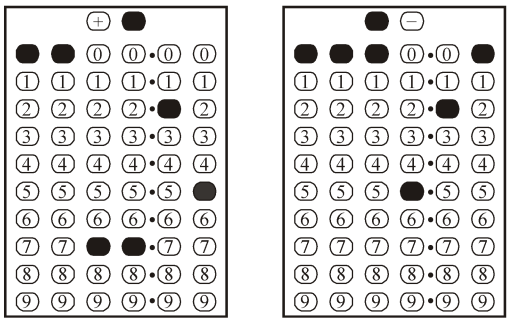
18. Which of the following is only correct match :
(A) A, P, II (B) C, Q, III (C) D, S, IV (D) B, Q, I

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SECTION-II: (Integer Second Decimal)

- This section contains **3** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

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1. A metal 'M' having atomic mass 31.25 crystallizes in cubic close packing and it shows " Schottky defects". If the edge length of the cubic lattice is 500 pm and density of the metal is 1.6075 gm/ml then calculate number of moles of metal atom 'M' missing per litre of the crystal.

[Given : 1 amu = 1.67×10^{-24} gm]

2. If truncated octahedron has x corners y square faces & z hexagon then
z - x/y = ?

3. If NaCl is dopped with 10^{-3} mol % $SrCl_2$, and the numbers of cation vacancies per mole of NaCl is $x \times 10^{18}$, the value of 'x' is ($N_A = 6 \times 10^{23}$

ANSWER KEY

SECTION–I : (i) Only One option correct Type 3(–1)

1.	**Ans. (C)**	**2.**	**Ans. (A)**	**3.**	**Ans. (B)**	**4.**	**Ans. (C)**
5.	**Ans. (C)**	**6.**	**Ans. (C)**	**7.**	**Ans. (D)**	**8.**	**Ans. (A)**

(ii) One or more options correct Type 4(–1)

9.	**Ans. (C)**	**10.**	**Ans. (A), (D)**	**11.**	**Ans. (A,C,D)**	**12.**	**Ans. (B,C)**

(iii) Paragraph Typ 3(0)

13.	**Ans. (B)**	**14.**	**Ans. (A)**

(iv) Matching List Type 4(0)

15.	**Ans. (A)**

(v) Table Type

16.	**Ans. (D)**	**17.**	**Ans. (B)**	**18.**	**Ans. (D)**

SECTION-II: (Integer Second Decimal)

1.	**Ans. (2.00)**	**2.**	**Ans. (4.00)**	**3.**	**Ans. 6.00**

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